Higher National Diploma in Engineering (C \ E \ M BSE)

First Year, First Semester Examination – 2020

MA 1101: Engineering Mathematics

Question 1 (Matrix Algebra)

Matrix A is defined as:

$$A = \begin{pmatrix} 1 & 0 & -1 \\ 1 & 2 & 1 \\ 2 & 2 & 3 \end{pmatrix}$$

a. Find the **eigenvalues** and the corresponding **eigenvectors** of the matrix A. b. Find the **Modal Matrix** and its inverse. c. **Diagonalize** matrix A.

(10+10+05 Marks)

Question 2 (Series and Convergence)

a. Show that if a series $\sum u_n$ is **convergent** then $\lim_{n\to\infty} u_n = 0$. b. Test the **convergence** of the following series: i. $\sum(\sqrt[3]{n^3+1} - n)$ ii. $\sum_1^\infty \frac{n!2^n}{n^n}$ c. For what value of x does the series $\sum(-1)^n(4x+1)^n$ **converge absolutely**?

(10+10+05 Marks)

Question 3 (Partial Differentiation)

a. Investigate the **continuity** of the following functions: i. $f(x,y) = \frac{x^2y^2}{x^2+y^2}$ when $(x,y) \neq (0,0)$ and $f(0,0) = 0$. ii. $f(x,y) = \frac{xy}{x^2+y^2}$ when $(x,y) \neq (0,0)$ and $f(0,0) = 0$. b. If $f = \ln(x^2 + y^2)$, then prove that $\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x}$. c. If $u(x,y) = x^2 + y^2$, then prove that $(\frac{\partial u}{\partial x} - \frac{\partial u}{\partial y})^2 = 4(1 - \frac{\partial u}{\partial x} - \frac{\partial u}{\partial y})$.

(10+10+05 Marks)

Question 4 (Improper Integrals)

a. Test the **convergence** of the improper integral: i. $\int_1^\infty \frac{x}{1+x^4} dx$ ii. $\int_{-1}^1 \frac{dx}{x^2}$ b. Evaluate the following **improper integral**, if it exists: i. $I = \int_0^\infty \frac{dx}{a^2+x^2}$, where $a > 0$ ii. $I = \int_0^3 \frac{dx}{\sqrt{9-x^2}}$

(05+06+07+07 Marks)

Question 5 (Multiple Integrals)

a. **Evaluate**:

$$\int_0^5 \int_0^{x^2} x(x^2 + y^2)\, dx\, dy$$

(Note: The integration limits in the source appear unconventional for a standard double integral setup, possibly $\int_0^5 \int_0^{x^2}$ *or a typo for* $dy\, dx$*)*. b. Show that the **area** between the parabolas $y^2 = 4ax$ and $x^2 = 4ay$ is $\frac{16}{3}a^2$. c. Show that, by **triple integration**, the **volume** of the sphere $x^2 + y^2 + z^2 = a^2$ is $\frac{4\pi a^3}{3}$.

(05+10+10 Marks)